CUSTOMER GUIDE

KEYCHAIN KREATURES

A little machine with a personality

An ESP32-S3 virtual-pet and handheld-game hardware project. The current design uses a repairable through-hole main board and a separate power board.

Current design at a glance

Main board: C.6, 96 × 105 × 1.6 mm. Power board: P.4, 50 × 50 × 1.6 mm. The boards require separate case supports; their mounting patterns do not match.

A 280 × 240 landscape display, nine controls, microSD storage, speaker output, vibration, infrared transmitter/receiver and RGB indication are part of the hardware design.

Development status

This is an engineering prototype, not a tested or certified finished toy. Physical fit, battery and case selection, power qualification and firmware are still being developed. The planned virtual pet, games, trading and update experience are not implemented or validated by this hardware release.

What customers can expect next

Availability and final specifications will be announced after prototype testing. This preview does not include a consumer assembly procedure or authorize connecting a battery to an unqualified board.

02 / KEYCHAIN KREATURES

Before a prototype is powered

Keep the two boards distinct

The main board and power board have different assembly requirements. The power module includes charging, protection and regulated outputs; it requires qualified assembly and independent testing before connection to the main board.

Power precautions

Never connect raw battery voltage directly to the main power input. Do not bypass the battery protection or combine programming USB and external power without the exact module’s reverse-feed behavior being qualified.

Neither speaker output is ground. Battery selection, temperature sensing and charging behavior require a documented, tested configuration.

Firmware and compatibility

Only firmware matched to the current ESP32-S3 hardware should be used. Historical sketches for earlier controllers and different peripheral chips are not compatible factory tests.

Radio performance, operating temperatures and simultaneous-load capacity have not been established for a finished product.

Support and updates

For current availability and documentation, visit the Keychain Kreatures listing on roddy.world. For prototype questions, email support@roddy.world and include the board revision. Do not send passwords or payment information.